

Marian College Kuttikkanam Autonomous

Initial Summative Assessment (ISA)

241DSCUBC104 Discrete Mathematics

Question Paper C

Time: 45 Minutes

Total Marks: 15

1. Express the following English sentences using symbols and logical connectives;

- i) "If the battery is charged and the device is powered on, then the Wi-Fi is connected."
- ii) "The battery is not charged or the Wi-Fi is connected, and the device is powered on."
- iii) "It is not true that if the device is powered on, the Wi-Fi is connected, if and only if the device is on but the Wi-Fi is not connected."

(CO1: 5 Marks)

2. Scenario:

A toy factory produces three types of toys: Cars (A), Dolls (B), and Trucks (C). The production constraints are as follows:

- A total of 500 toys are produced.
- The number of Dolls produced is 30 more than the number of Cars produced.
- The total production of Cars and Trucks is 350.

Question:

Formulate a system of linear equations from the given scenario and determine how many toys were sold for each category.

(CO5; 10 Marks)

Course Outcomes

CO1: Apply rules of inferences for propositional logic to solve logical and mathematical problems.

CO5: Solve the system of linear equations.